

Marketing Analytics Problem Assignment: Part 2

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Introduction:

This report examines consumer behaviour by using the dataset of Purchasing_data.csv, specifically the variables that affect the consumer propensity to utilize Buy Now, Pay Later (BNPL) services. Two regression models were built: Model 1, a simple linear regression, and Model 2, a multiple regression with three predictors. An initial exploratory data analysis revealed that credit card ownership, monthly credit card balance, and savings levels were marginally correlated, which means that the selected financial variables can determine the BNPL usage. The evidence of regression indicates that the debt levels positively affect BNPL likelihood, whereas a larger number of savings and credit cards negatively affect it. Diagnostic testing showed that there were slight violations of normality and homoscedasticity, but the model can be used with strong corrections. These lessons can be used by financial institutions that want to have a deeper understanding of the BNPL adoption and improve their approach to targeting their customers.

Exploratory Analysis

Dependent Variable (DV)

q11_1_num - Likelihood to Use BNPL (1–5 scale)

Independent Variables (IVs)

- **q02_num** – Number of credit cards
- **q03_num** – Outstanding non-mortgage debt
- **d16_num** – Savings level

Correlation Matrix (Rounded 3 decimals)

	q11_1_num	q02_num	q03_num	d16_num
q11_1_num	1.000	-0.112	0.231	-0.071
q02_num	-0.112	1.000	0.209	0.115
q03_num	0.231	0.209	1.000	0.150
d16_num	-0.071	0.115	0.150	1.000

Insights:

- BNPL likelihood increases as debt increases.
- BNPL decreases slightly as the credit card count increases.

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- Higher savings reduce BNPL usage, but weakly.

These trends support using linear regression, not non-linear models.

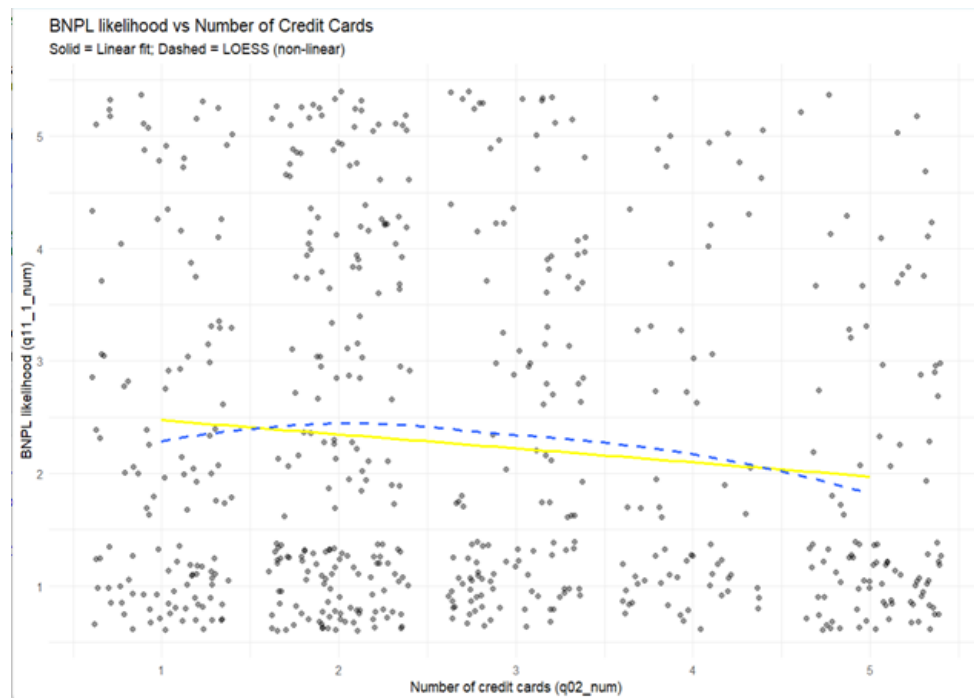


Figure 1. Scatterplot of BNPL likelihood vs Number of Credit Cards

BNPL vs number of credit cards (q02_num). The LOESS line is a bit curved, although it remains near the downward trend, which is smooth like a linear line. No hard non-linear trend is present there is an acceptable linear correlation.



Figure 2. Scatterplot of BNBL likelihood vs Non-mortgage Debt Level

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Linear as well as LOESS lines depict an evident upward tendency: the more debt one possesses, the more they are interested in the use of BNPL. Once again, curvature is not too severe, and thus a linear model can be used.

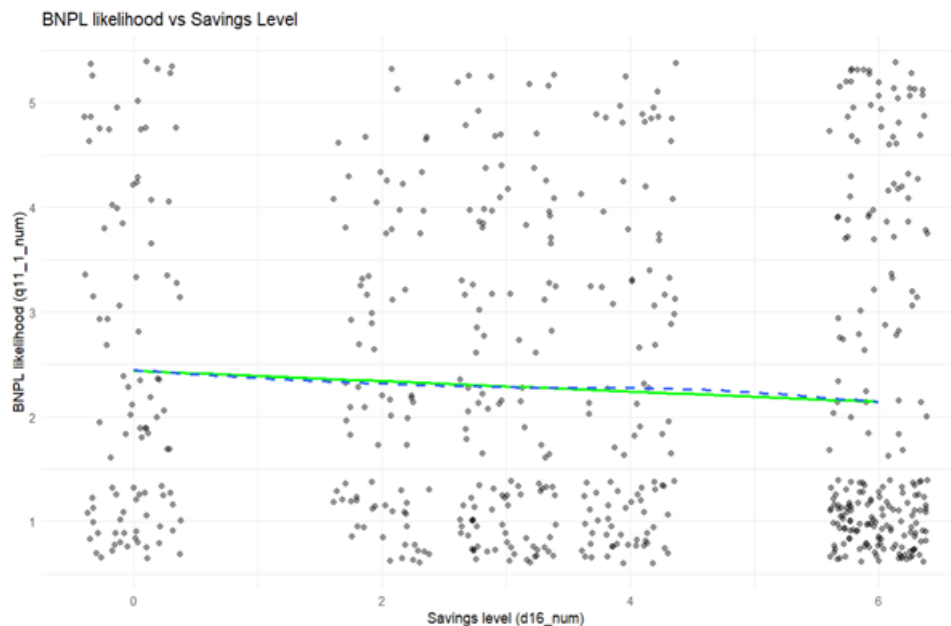


Fig 3. Scatterplot of BNPL likelihood vs Savings Level

The downward slope in both lines is not that significant; the more a respondent is saving, the less likely they are to use BNPL. Trends are quite linear in nature - linear specification is warranted.

Model 1 – Simple Linear Regression

Model 1 formula:

$$\text{BNPL likelihood (q11_1_num)} = \beta_0 + \beta_1(\text{q02_num}) + \epsilon$$

statistics:

- Intercept: 2.598 ($p < 0.001$)
- q02_num coefficient: -0.125
- Residual standard error: 1.492
- $R^2 = 0.0126$, Adjusted $R^2 = 0.0110$
- F-statistic = 7.61, $p = 0.006$

Key Findings:

- Every additional credit card is correlated with an average reduction in the BNPL probability by 0.125 points (on a 1-5 scale), other factors being equal.
- The impact is statistically significant but insignificant in size.
- One can only explain 1.3% of BNPL predictability with the model, meaning that credit card count is weakly predictive.
- Since the exploratory plots indicated that there is more than a single financial measure that could be valuable, we transitioned to multiple regression.

Model 2 – Multiple Linear Regression

Model 2 formula:

$$\text{BNPL likelihood} = \beta_0 + \beta_1(\text{q02_num}) + \beta_2(\text{q03_num}) + \beta_3(\text{d16_num}) + \varepsilon$$

statistics:

Intercept: 2.319 ($p < 0.001$)

- Residual standard error: 1.436
- $R^2 = 0.089$, Adjusted $R^2 = 0.084$
- F-statistic = 19.33 on 3 and 593 DF, p-value: 5.694e-12
- Error metrics:
- Model 1: RMSE = 1.490, MAE = 1.316
- Model 2: RMSE = 1.431, MAE = 1.222

Key Findings:

- Positive coefficients indicate a positive relationship between two variables, while negative coefficients show a negative relationship between two variables.
- q03_num (Debt level) -With each increment of the debt category, BNPL probability increases by an average of 0.18 points. BNPL is more likely to be used by consumers who have outstanding balances.
- q02_num (Number of credit cards) -With each extra credit card, the likelihood of BNPL on average decreases by 0.18 points, all else being equal, with consideration of debt and savings.

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Individuals who hold numerous cards might already have additional credit opportunities and might not use BNPL so much.

- d16 num (Savings) – With every increase in savings bracket, one will decrease BNPL probability by an average of 0.07 points, indicating that less-savings customers have higher BNPL probabilities.

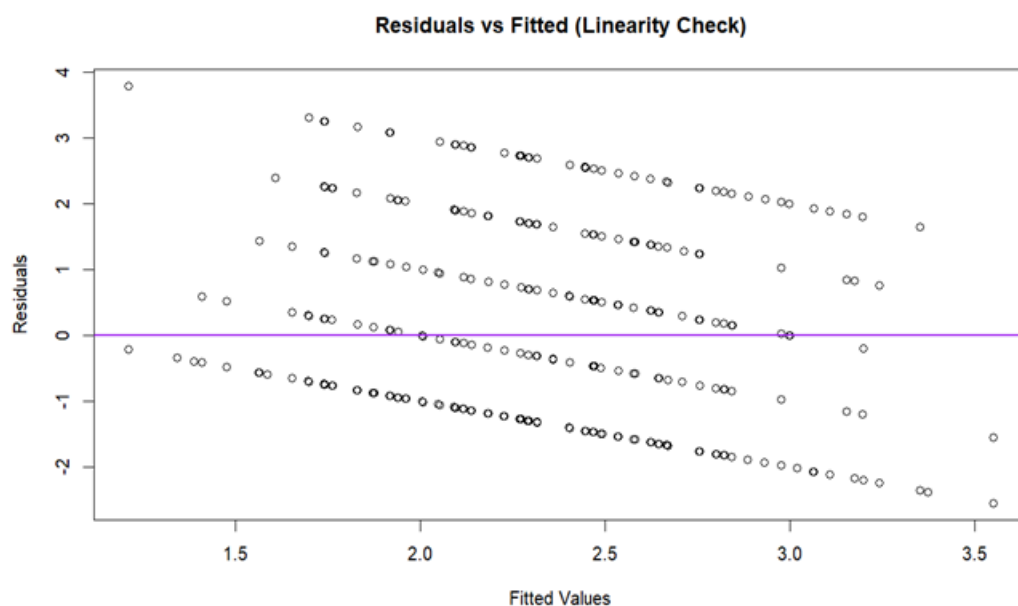
Model fit comparison:

The R^2 of Model 2 (0.089) is bigger than the R^2 of Model 1 (0.013). The multiple models explain seven times more than the simple model, though it is still modest.

The values of RMSE and MAE are lower in Model 2, which implies that its predictions are a little more accurate. Model 2 is preferred since it is more meaningful and has a better performance on all metrics than Model 1.

Diagnostic Testing:

Plot: Fitted vs. Residuals (Check Fitted vs. Residuals) There are no strong curves or systematic trends around the zero line, and horizontal bands are observed in the plot as the response is ordinal in nature.

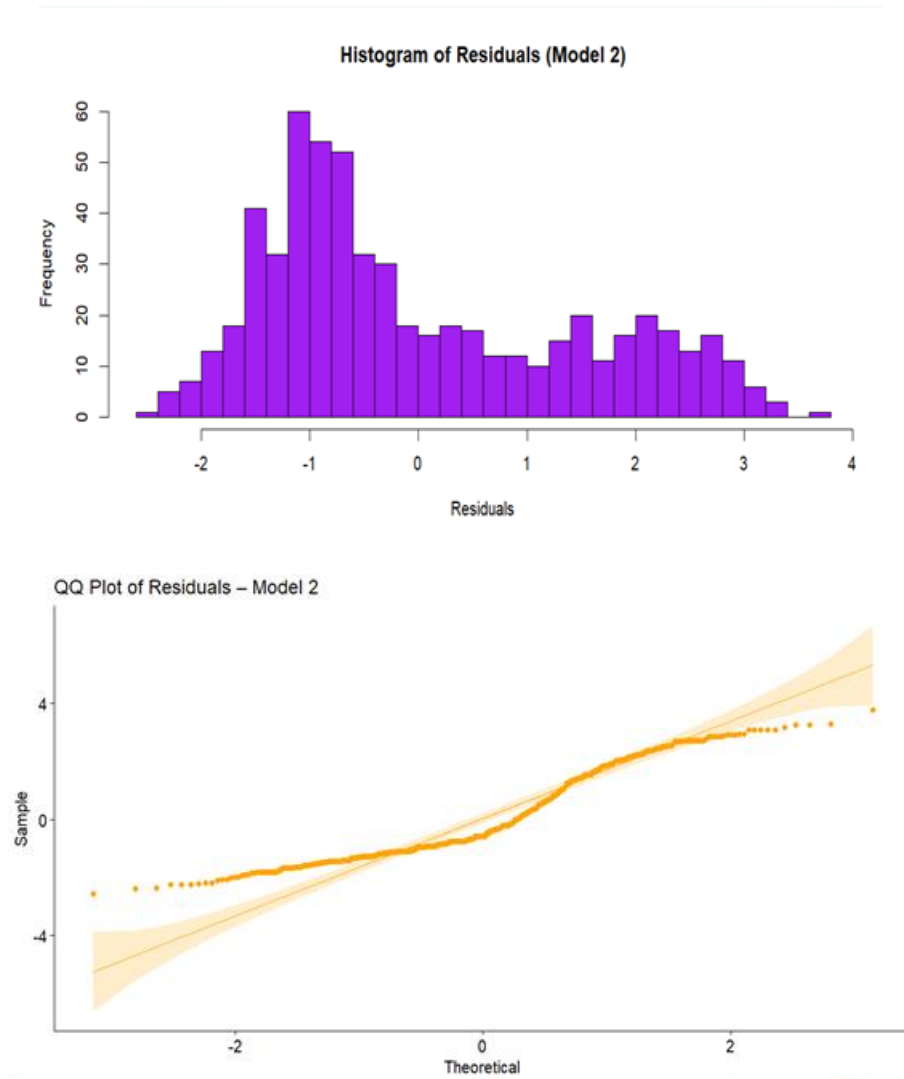


Interpretation: The linearity assumption is in part met and with minor nonconformities. They are not harsh to nullify the model, but they suggest that non-linear models might be able to represent the relationships in future analysis in a more accurate way.

Normality of Residuals

The QQ plot shows the residuals mostly following the straight line in the middle range but deviating at both tails.

The histogram confirms mild right-skewness.



Homoscedasticity (Breusch–Pagan Test)

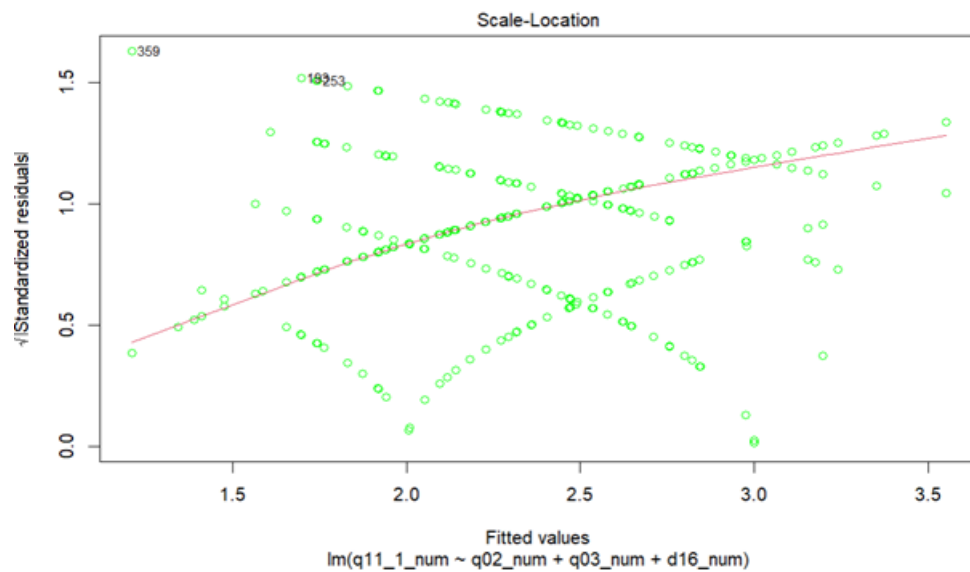
BP = 27.24, df = 3, p-value = 5.243e-06

Because $p < 0.05$, we reject the null hypothesis of constant variance.

The model exhibits heteroscedasticity.

This is confirmed visually in the scale-location plot, where residuals widen slightly at higher fitted values.

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Interpretation:

The equal variance assumption is not met, and that is, the standard errors of the basic regression could be biased.

This is the reason why you used HC3 robust standard errors, which correct the heteroscedasticity and render the p-values credible.

Multicollinearity

VIF values were:

- $q02_num$ - 1.05
- $q03_num$ - 1.06
- $d16_num$ - 1.03

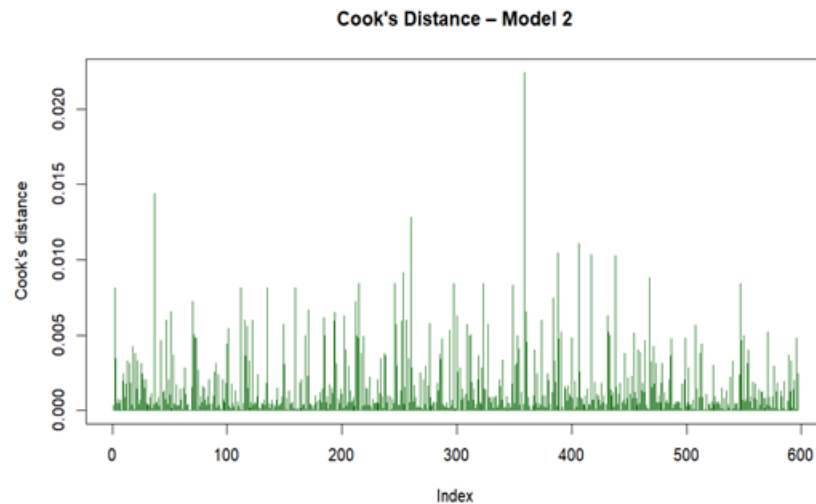
Interpretation:

VIF values are extremely low, indicating no multicollinearity.

Thus, the predictors measure distinct concepts and do not overlap.

5. Influence and Outliers (Cook's Distance)

The Cook's Distance plot showed a few points exceeding the threshold $4/n$, but none near the severe cutoff of 1.0.



Interpretation:

While some observations have a mild influence, none pose a serious threat to model stability. The regression results remain trustworthy, and removing these points is not necessary.

Overall Diagnostic Summary

Linearity	Slight deviations	Model is mostly linear; relationship is acceptable.
Normality of residuals	Mild skewness	Acceptable for ordinal Likert-scale outcomes.
Homoscedasticity	Violated (BP $p < .001$)	Heteroscedasticity present robust SE needed.
Multicollinearity	No issue (VIF ~ 1.05)	Predictors are independent.
Influential Points	No strong outliers	Model remains stable.

Final Interpretation: According to the diagnostic tests, the model deviates slightly against the ideal assumptions, which are mostly non-normal residuals and heteroscedasticity, but the application of HC3 robust standard errors resolves the problem and yields valid statistical inference. The model is reliable, explainable and suitable to describe the trends in the likelihood of BNPL.

Model Comparison

R ²	0.0126	0.0890	Model 2 is better
RMSE	1.49	1.431	Lower = better
MAE	1.316	1.222	Lower = better

Key Findings:

Model 2 has the highest accuracy when it comes to prediction, with lower RMSE and MAE than Model 1. The improvement is not large but indicates that the incorporation of debt and savings has a significant positive effect on the model.

Business Implications

1. Debt level is the strongest predictor.

- Customers with higher loans are more likely to adopt BNPL services.
- Marketing teams can target debt-heavy customers with BNPL promotions.

2. Customers with high savings or many credit cards are less interested in BNPL.

- BNPL is not attractive to financially secure or multi-card users.
- Resources should not be wasted on these groups.

3. Behavioural segmentation is essential.

- Younger or financially stretched groups are more responsive to BNPL offers.

4. Marketing Budget Optimisation

- Redirect ads toward high-debt and low-saving segments.
- Avoid blanket campaigns - use targeted messaging.

Limitations:

The dependent variable is ordinal, yet linear regression is done as numeric. There is a low R² characteristic of human behaviour datasets. SE had to be strong to counter heteroscedasticity. There was massive missingness in the dataset, which decreased the sample size. There may be non-linear tendencies that were not captured by this model.

Improvement Recommendation:

See nonlinear models, including decision trees and polynomial regression. Add age, education, and marketing exposure to more effective predictive variables. Ordinal logistic regression (more suitable in the case of Likert-scale DV). Test other predictors with regularized models (LASSO/ Ridge).

Conclusion:

This discussion has shown that BNPL likelihood is largely determined by the debt levels, with savings and credit cards ownership decreasing the interest of BNPL. Multiple regression (Model 2) is superior to simple regression, but still, it only explains a moderate level of variance, which is common with behavioural survey data. According to business knowledge, there is a high value when it comes to targeting financially constrained consumers in advancing BNPL promotions. More behavioural variables should be studied with non-linear models and other variables that would help enhance the predictive performance in future.